

Frequency Bias Signal Malfunctions

Symptoms

- Frequency bias signal **not** available from customer interface box.

How To Use This Tree

This symptom tree can be used to troubleshoot engine symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Customer Interface Box Wiring	
	STEP 1A. Check Generator Speed/Load Governing Bias Supply Wire	Less than 10 ohms resistance?
	STEP 1B. Check Generator Speed/Load Governing Bias Return Wire	Less than 10 ohms resistance?
	STEP 1C. Check Generator Speed/Load Governing Bias Signal Wire	Less than 10 ohms resistance?
STEP 2.	Check Engine Harness to Customer Interface Box Cable	
	STEP 2A. Check Generator Speed/Load Governing Supply and Signal Wires	Less than 10 ohms resistance?
	STEP 2B. Check Generator Speed/Load Governing Return and Signal Wires	Less than 10 ohms resistance?

STEP 1. Check Customer Interface Box Wiring

STEP 1A. Check Generator Speed/Load Governing Bias Supply Wire

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.

Action	Specification/Repair	Next Step
Check the generator speed/load governing bias supply wire. • Place one test lead on the generator speed/load governing bias 5 volt supply (sensor supply 4) pin in connector C3. • Place the other test lead on the generator speed/load governing bias 5 volt supply (sensor supply 4) terminal on the X4 connector.	Less than 10 ohms resistance? YES	1B
	Less than 10 ohms resistance? NO Repair : Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 1B. Check Generator Speed/Load Governing Bias Return Wire

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.

Action	Specification/Repair	Next Step
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Check the generator speed/load governing bias return wire. - Place one test lead on the generator speed/load governing bias return (sensor return 4) pin in connector C3. - Place the other test lead on the generator speed/load governing bias return (sensor return 4) terminal on the X4 connector.	Less than 10 ohms resistance? YES	1C
	Less than 10 ohms resistance? NO Repair : Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 1C. Check Generator Speed/Load Governing Bias Signal Wire

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.

Action	Specification/Repair	Next Step
Check the generator speed/load governing bias signal wire. - Place one test lead on the generator speed/load governing bias signal pin in connector C3. - Place the other test lead on the generator speed/load governing bias signal terminal on the X4 connector.	Less than 10 ohms resistance? YES	2A
	Less than 10 ohms resistance? NO Repair : Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 2. Check Engine Harness to Customer Interface Box Cable

STEP 2A. Check Generator Speed/Load Governing Supply and Signal Wires

Conditions :

- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.
- Disconnect customer interface box to engine harness cable connector C10 from the engine harness.

Action	Specification/Repair	Next Step
Check generator speed/load governing bias supply and signal wires. • Place a jumper between the generator speed/load governing bias 5 volt supply (sensor supply 4) pin and the generator speed/load governing bias signal pin in the C10 connector. • Place one test lead in the generator speed/load governing bias 5 volt supply (sensor supply 4) pin of the C3 connector. • Place the other test lead in the generator speed/load governing bias signal pin of the C3 connector.	Less than 10 ohms resistance? YES	2B
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 2B. Check Generator Speed/Load Governing Bias Return and Signal Wires

Conditions :

- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.
- Disconnect customer interface box to engine harness cable connector C10 from the engine harness.

Action	Specification/Repair	Next Step
Check generator speed/load governing bias supply and signal wires. • Place a jumper between the generator speed/load governing bias return (sensor return 4) pin and the generator speed/load governing bias signal pin in the C10 connector. • Place one test lead in the generator speed/load governing bias return (sensor return 4) pin of the C3 connector. • Place the other test lead in the generator speed/load governing bias signal pin of the C3 connector.	Less than 10 ohms resistance? YES Repair : Refer to Section TF in the Troubleshooting and Repair Manual, QSK19 CM850 Modular Common Rail System Series Engines, Bulletin 4021493 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021493.html) or refer to the OEM Service Manual for potentiometer repair instructions.	Repair complete.
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.